Norton's Theorem

Video 8

Electrical Science

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Norton's Theorem

- Norton's Theorem
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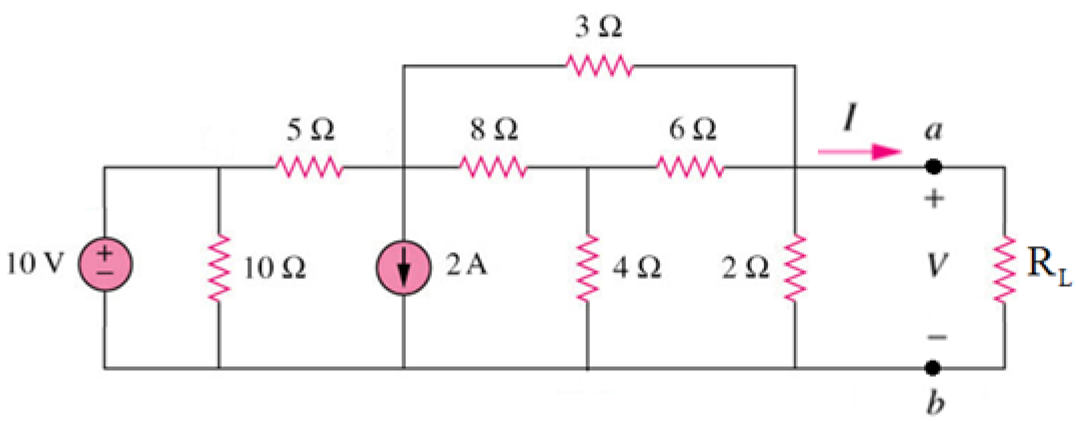
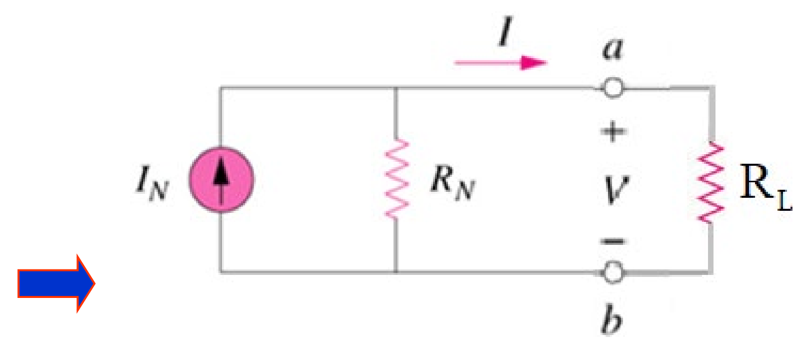
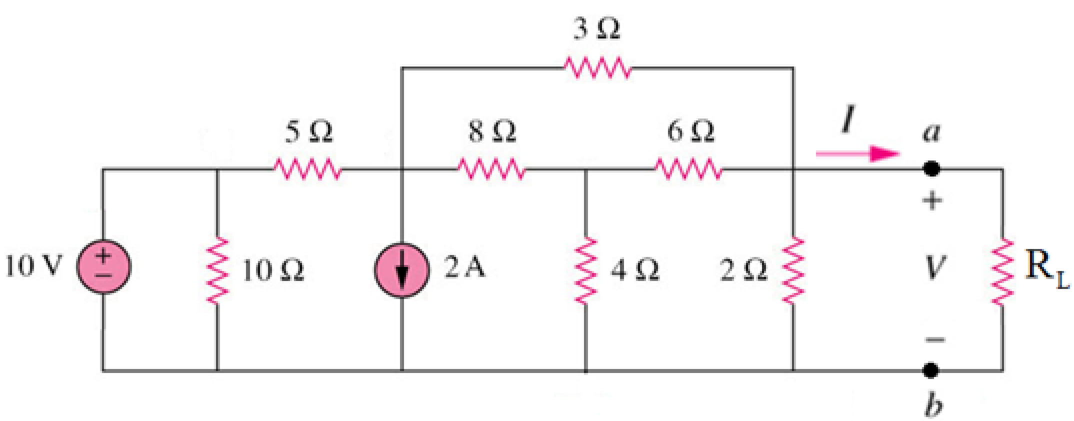
- Wheatstone Bridge

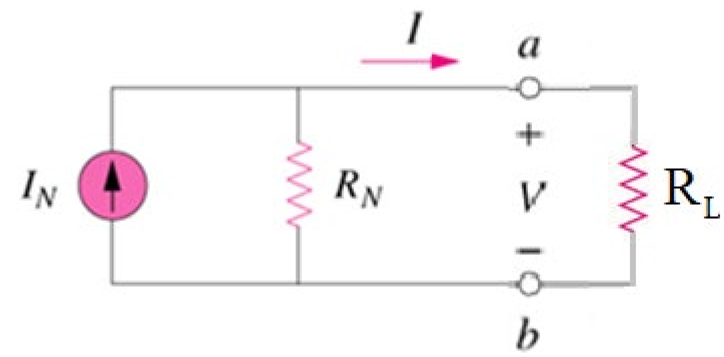




Norton's Theorem

It states that a linear two-terminal circuit can be replaced by an equivalent circuit of a current source I_N in parallel with a resistor R_N.



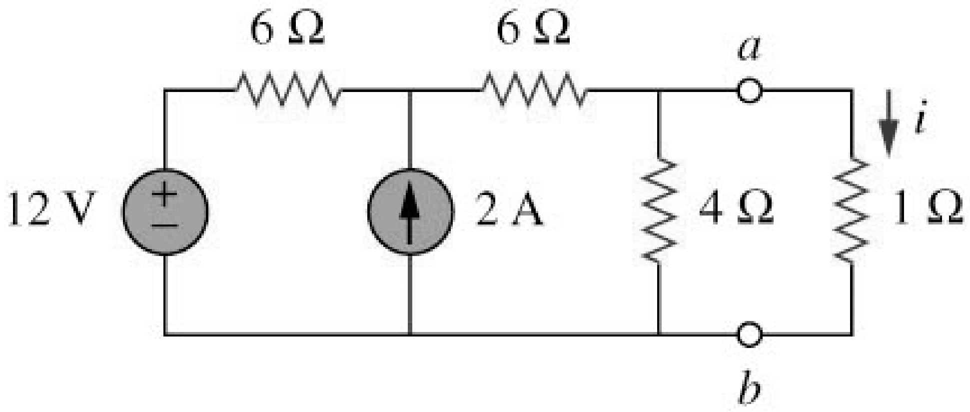


- I_N is the short circuit current through the terminals.
- R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.
- Independent sources are turned off by replacing a Voltage source by a short circuit and current source by open circuit.

Norton’s Theorem

Example 1

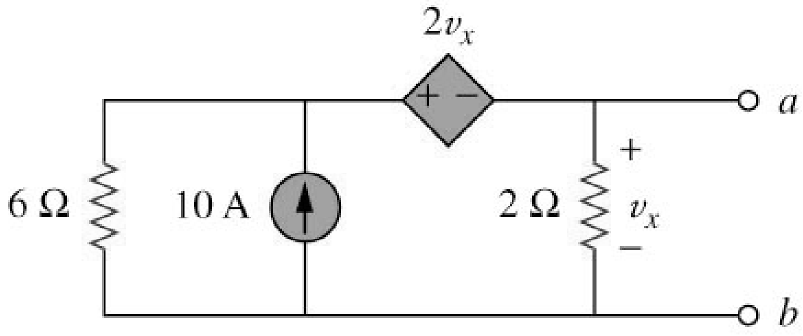
Using Norton’s theorem, find the equivalent circuit to the left of the terminals in the circuit shown below. Hence find i.



Norton's Theorem with Dependent Source

Example 2

Find the Norton equivalent circuit of the circuit shown below.



Wheatstone Bridge

Example 3

The Wheatstone bridge circuit shown below is used to measure the resistance of a strain gauge. The bridge is balanced when the adjustable resistor is 21.3 ohms. Find the resistance of the strain gauge when the bridge is balanced?